

RESEARCH PROJECT PORTFOLIO

CAMO

Class-Aware Minority-Optimized Ensemble
for Robust Language Model Evaluation on Imbalanced Data

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📄 Under Review • Applied
Intelligence (Q2 Journal)

🏆 State-of-the-Art Macro F1

🧠 8 LLMs • 2 Datasets • 7
Baselines

01 | PROJECT OVERVIEW

CAMO (Class-Aware Minority-Optimized) is a novel ensemble framework developed to solve one of the most persistent challenges in real-world AI classification — class imbalance. Traditional ensemble methods systematically marginalize minority class predictions by favoring the majority, leading to poor macro F1 performance. CAMO introduces a hierarchical, uncertainty-aware decision architecture that dynamically boosts underrepresented classes without sacrificing overall accuracy.

The framework was validated across two highly imbalanced, domain-specific benchmarks using eight language models under both zero-shot and fine-tuned conditions, benchmarked against seven established ensemble algorithms. CAMO consistently achieved the highest strict macro F1 score among all competing methods with fine-tuned models.

02 | PROJECT DETAILS

Project Type	Research Paper — Novel Ensemble Method for Imbalanced NLP Classification
Domain	NLP • Large Language Models • Ensemble Learning • Imbalanced Classification
Publication Status	Under Review — Applied Intelligence Journal (Springer, Q2)
Core Technologies	Python • PyTorch • HuggingFace Transformers • LoRA / QLoRA • LLaMA • Mistral • DeepSeek • Phi-3
Affiliation	MSA University (Egypt) • Qatar University (Qatar)

03 | KEY RESULTS & METRICS

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Language Models LLMs & SLMs evaluated	Ensemble Baselines All outperformed by CAMO	Datasets Imbalanced benchmarks	Best Result Strict Macro F1 Score
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Performance on BEA 2025 Dataset (Fine-Tuned Models — Strict F1)

Model	Strict F1	Strict Accuracy	Lenient F1	Lenient Accuracy
Phi-3 (3.8B)	85.6%	93.2%	93.0%	96.5%
Llama-3.2 (1B)	75.9%	88.9%	87.9%	94.1%
Qwen3 (0.6B)	70.2%	86.4%	84.1%	92.6%
DeepSeek-R1 (1.5B)	57.6%	82.5%	76.3%	90.1%
Mathstral-7B	80.1%	91.0%	91.6%	95.8%
DeepSeek-R1-8B	78.4%	89.8%	89.8%	94.9%
Qwen3-8B (8B)	74.0%	88.9%	88.9%	94.5%
Falcon3 (1B)	51.0%	81.1%	70.0%	88.7%

Table 1 Performance comparison of ensemble methods across fine-tuned language models (percentages) on BEA DATASET. Each cell shows: **Strict F1 / Strict Accuracy / Lenient F1 / Lenient Accuracy**

Model	Ensemble Method							
	Maj. Vote	Conf. Weight	Dyn. Thr.	Class Bal.	Unc. Aware	Meta Ens.	MSA Base.	CAMO
Phi-3 (3.8B)	77.6	76.9	83.7	77.3	78.6	78.6	82.6	85.6
	91.4	91.3	92.9	91.3	90.2	91.6	92.4	93.2
	92.0	92.3	92.8	92.0	91.5	92.0	92.0	93.0
	95.9	96.0	96.3	95.9	95.8	95.9	95.9	96.5
DeepSeek-R1 (1.5B)	51.9	54.4	55.0	54.4	56.0	54.2	55.5	57.6
	83.3	83.7	83.6	83.7	79.8	83.6	83.9	82.5
	78.4	78.6	78.0	78.7	69.5	78.4	78.8	76.3
	90.6	90.6	90.3	90.8	88.6	90.6	90.8	90.1
Qwen3 (0.6B)	62.5	62.1	63.0	62.9	63.4	62.2	63.7	70.2
	86.4	86.3	86.3	86.2	81.7	86.3	86.3	86.4
	85.0	84.5	84.8	84.7	79.7	84.5	84.7	84.1
	92.9	92.6	92.8	92.8	91.4	92.6	92.8	92.6
Llama-3.2 (1B)	66.4	64.8	66.3	67.4	69.4	66.7	68.3	75.9
	87.6	87.8	87.8	88.0	84.9	87.9	87.8	88.9
	86.5	86.4	86.4	87.1	84.5	86.8	86.5	87.9
	93.3	93.2	93.2	93.6	93.1	93.5	93.3	94.1
Falcon3 (1B)	48.8	48.8	49.0	49.1	42.2	48.8	48.8	51.0
	82.4	82.5	82.5	82.5	75.6	82.4	82.4	81.1
	75.4	75.4	75.6	75.8	49.8	75.4	75.4	70.0
	89.9	89.9	89.9	90.1	84.5	89.9	89.9	88.7
Mathstral-7B (7B)	75.2	76.5	79.2	75.6	78.0	77.1	79.0	80.1
	90.6	91.0	91.6	90.8	89.9	91.2	91.2	91.0
	91.2	91.8	91.9	90.9	91.1	92.0	90.8	91.6
	95.5	95.8	95.9	95.4	95.6	95.9	95.4	95.8
Qwen3-8B (8B)	65.3	65.1	69.1	66.2	66.9	65.3	70.1	74.0
	88.0	87.9	88.3	88.2	85.5	88.0	88.6	88.9
	89.4	89.1	89.2	89.4	84.4	89.4	89.4	88.9
	94.7	94.5	94.5	94.7	92.9	94.7	94.7	94.5
DeepSeek-R1-8B (8B)	72.4	76.0	78.0	74.7	74.3	74.5	77.2	78.4
	89.5	90.3	90.8	89.9	87.6	89.9	90.2	89.8
	90.0	90.6	90.6	89.7	88.6	90.2	89.9	89.8
	94.9	95.2	95.2	94.8	94.5	95.1	94.9	94.9

Performance on DIAR-AI Emotion Dataset

Model	Accuracy	Precision	Recall	Macro F1
Llama-8B (CAMO)	92.75%	87.94%	90.15%	88.70%
Llama-1B (CAMO)	92.45%	87.02%	92.33%	88.84%
GPT-4o-DPO (Baseline)	93.10%	90.80%	87.09%	87.90%

04 | TECHNICAL APPROACH

The Problem

Class imbalance causes traditional ensemble methods to systematically favor majority classes. In real-world datasets — such as educational assessment (7% minority class) and emotion recognition (3% minority class) — this means the most important and nuanced predictions are consistently wrong.

CAMO's Solution: Hierarchical Decision Architecture

CAMO introduces a 7-stage hierarchical decision pipeline that processes ensemble predictions through progressive checks before returning a final class. Each stage is mathematically formalized with dynamic thresholds:

- Stage 1 — Unanimity Check: Returns unanimous prediction when all models agree
- Stage 2 — Strong Minority Consensus: Boosts minority class when multiple models agree with high confidence
- Stage 3 — Isolated High-Confidence Minority Vote: Preserves single high-confidence minority predictions
- Stage 4 — Uncertainty-Triggered Minority Prioritization: Activates minority preference under high model uncertainty
- Stage 5 — Confidence-Weighted Aggregation with Dynamic Boosting (1.8–2.5x for minority classes)
- Stage 6 — Minority Vote Under Competitive Dominance: Resolves ties in favor of minority classes
- Stage 7 — Standard Majority Rule: Falls back to standard aggregation when no conditions are met

Fine-Tuning Strategy

All language models were fine-tuned using parameter-efficient methods to ensure reproducibility and scalability:

- LoRA (Low-Rank Adaptation) with rank 32 and alpha 64 for efficient parameter adaptation
- 4-bit quantization (QLoRA) for memory-efficient training on resource-constrained hardware
- Multi-seed training (5 seeds) for robust uncertainty estimation
- AdamW optimizer with cosine scheduling and gradient checkpointing
- Early stopping based on macro F1 score to prevent majority-class overfitting

05 | KEY CONTRIBUTIONS & IMPACT

Novel Ensemble Framework First domain-agnostic ensemble method with explicit minority class preservation, mathematically formalized with dynamic confidence thresholds and uncertainty-aware decision boundaries.	State-of-the-Art Results Outperformed 7 baseline ensemble methods including majority voting, confidence-weighted averaging, meta-ensembles, and uncertainty-aware methods across multiple architectures.
Comprehensive Benchmarking Evaluated across 8 language models (3 LLMs + 5 SLMs), 2 imbalanced datasets, and both zero-shot and fine-tuned settings — one of the most comprehensive comparative evaluations for imbalanced classification.	Real-World Applicability CAMO's minority-aware design directly addresses AI fairness concerns in high-stakes domains including fraud detection, rare disease diagnosis, educational AI, and emotion recognition.

06 | ABOUT THE RESEARCHER

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Education: Dual B.Sc. Computer Science — MSA University (Egypt) & University of Greenwich (UK), Expected 2026

Specialization: NLP, LLMs, Transformer Architectures, Deep Learning, Computer Vision, End-to-End ML Pipelines

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